

servo to the Arduino to determine its minimum and maximum positions, ensuring full range of motion.

Assemble the remaining components using screws, as shown in the figure. To complete the assembly, install

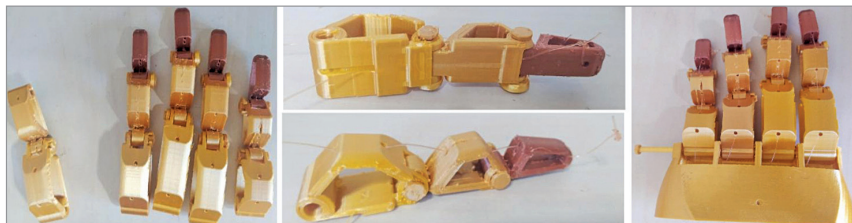


Fig. 6: Assembly of robot finger

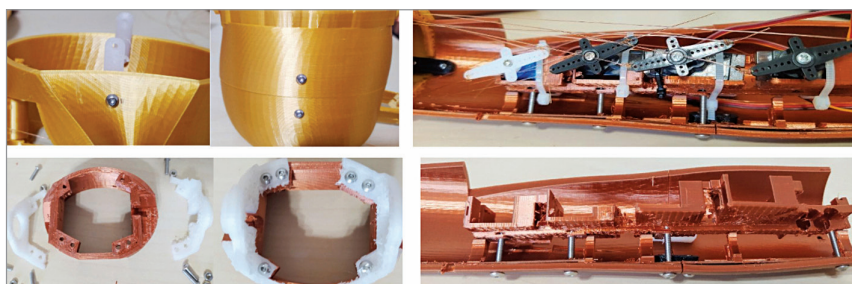


Fig. 7: Assembly of the arm wrist and hand

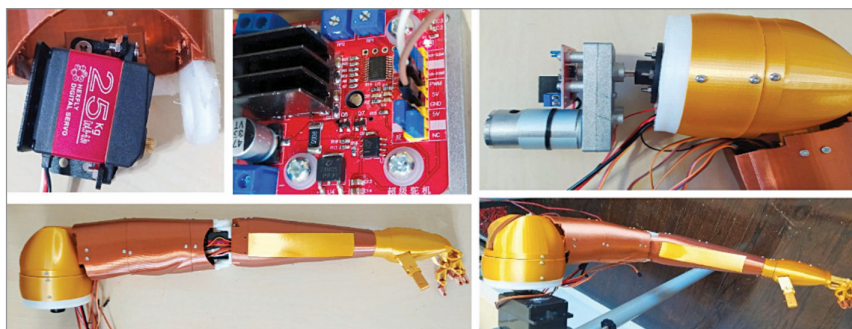


Fig. 8: Assembly of the servo motor and the complete robotic arm with all components integrated

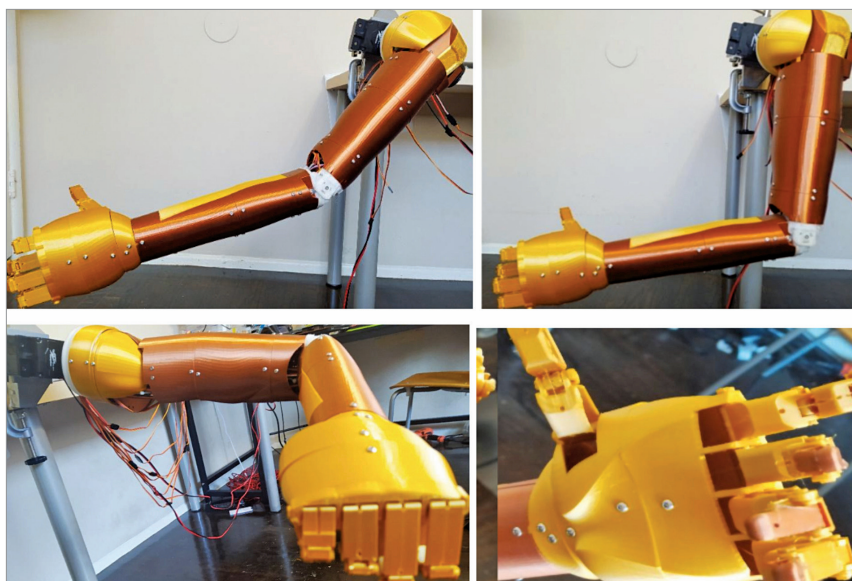


Fig. 9: Testing the robot arm movement with example code

## TO DOWNLOAD SOURCE CODE AND RESOURCES



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the second 180kg high-torque servo into the robotic arm. A bench vise can be used to hold the arm securely during this process. Always rotate each servo to its required position before fixing it in place, as identifying its full range of motion is essential for accurate programming. Fig. 8 shows the assembly of the servo motor and the complete robotic arm with all components integrated.

## Testing

After properly uploading the source code to the Arduino Uno and completing the assembly, the robotic arm is ready for testing.

For testing, power the Arduino with a 5V adaptor and supply the big servos using a 12V power source with at least 20A capacity. Keep in mind that small servos require lower voltage. Run the program; the arm will move to preset positions such as 90° and 180°, and the fingers will open and close according to the programmed sequence.

Once verified, the robotic arm is ready for further development. Functionality can be enhanced by adding custom control logic or integrating the system with ROS or other robotic control systems to perform more task-oriented operations. Fig. 9 shows a prototype of robot arm movement testing with example code.

**EFY note.** The Arduino, PCA9685, and all external servo power supplies must share a common ground. **EFY**



Kelvin Gonzalez Amador is an electrical engineer and the founder of Krobotics.